

pinecone-vector-db-image-serach

August 23, 2025

```
[1]: # Install dependencies
```

```
!pip install torch torchvision pillow transformers
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: torch in /usr/local/lib/python3.10/site-packages
(2.1.1)
Requirement already satisfied: torchvision in /usr/local/lib/python3.10/site-
packages (0.16.1)
Requirement already satisfied: pillow in /usr/local/lib/python3.10/site-packages
(10.1.0)
Requirement already satisfied: transformers in /usr/local/lib/python3.10/site-
packages (4.35.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.10/site-
packages (from torch) (3.13.1)
Requirement already satisfied: typing-extensions in
./usr/local/lib/python3.10/site-packages (from torch) (4.14.1)
Requirement already satisfied: sympy in /usr/local/lib/python3.10/site-packages
(from torch) (1.12)
Requirement already satisfied: networkx in /usr/local/lib/python3.10/site-
packages (from torch) (3.2.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.10/site-packages
(from torch) (3.1.2)
Requirement already satisfied: fsspec in /usr/local/lib/python3.10/site-packages
(from torch) (2023.10.0)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.1.105 in
/usr/local/lib/python3.10/site-packages (from torch) (12.1.105)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.1.105 in
/usr/local/lib/python3.10/site-packages (from torch) (12.1.105)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.1.105 in
/usr/local/lib/python3.10/site-packages (from torch) (12.1.105)
Requirement already satisfied: nvidia-cudnn-cu12==8.9.2.26 in
/usr/local/lib/python3.10/site-packages (from torch) (8.9.2.26)
Requirement already satisfied: nvidia-cublas-cu12==12.1.3.1 in
/usr/local/lib/python3.10/site-packages (from torch) (12.1.3.1)
Requirement already satisfied: nvidia-cufft-cu12==11.0.2.54 in
/usr/local/lib/python3.10/site-packages (from torch) (11.0.2.54)
Requirement already satisfied: nvidia-curand-cu12==10.3.2.106 in
/usr/local/lib/python3.10/site-packages (from torch) (10.3.2.106)
```

Requirement already satisfied: nvidia-cusolver-cu12==11.4.5.107 in /usr/local/lib/python3.10/site-packages (from torch) (11.4.5.107)

Requirement already satisfied: nvidia-cuspars-cu12==12.1.0.106 in /usr/local/lib/python3.10/site-packages (from torch) (12.1.0.106)

Requirement already satisfied: nvidia-nccl-cu12==2.18.1 in /usr/local/lib/python3.10/site-packages (from torch) (2.18.1)

Requirement already satisfied: nvidia-nvtx-cu12==12.1.105 in /usr/local/lib/python3.10/site-packages (from torch) (12.1.105)

Requirement already satisfied: triton==2.1.0 in /usr/local/lib/python3.10/site-packages (from torch) (2.1.0)

Requirement already satisfied: nvidia-nvjitlink-cu12 in /usr/local/lib/python3.10/site-packages (from nvidia-cusolver-cu12==11.4.5.107->torch) (12.2.140)

Requirement already satisfied: numpy in /usr/local/lib/python3.10/site-packages (from torchvision) (1.26.2)

Requirement already satisfied: requests in ./local/lib/python3.10/site-packages (from torchvision) (2.32.5)

Requirement already satisfied: huggingface-hub<1.0,>=0.16.4 in ./local/lib/python3.10/site-packages (from transformers) (0.34.3)

Requirement already satisfied: packaging>=20.0 in ./local/lib/python3.10/site-packages (from transformers) (24.2)

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.10/site-packages (from transformers) (6.0.1)

Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.10/site-packages (from transformers) (2023.10.3)

Requirement already satisfied: tokenizers<0.19,>=0.14 in /usr/local/lib/python3.10/site-packages (from transformers) (0.15.0)

Requirement already satisfied: safetensors>=0.3.1 in /usr/local/lib/python3.10/site-packages (from transformers) (0.4.1)

Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.10/site-packages (from transformers) (4.66.1)

Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in ./local/lib/python3.10/site-packages (from huggingface-hub<1.0,>=0.16.4->transformers) (1.1.5)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.10/site-packages (from jinja2->torch) (2.1.3)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.10/site-packages (from requests->torchvision) (3.3.2)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/site-packages (from requests->torchvision) (3.6)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/site-packages (from requests->torchvision) (1.26.18)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/site-packages (from requests->torchvision) (2023.11.17)

Requirement already satisfied: mpmath>=0.19 in /usr/local/lib/python3.10/site-packages (from sympy->torch) (1.3.0)

```
[2]: !pip uninstall -y pinecone-client
```

WARNING: Skipping pinecone-client as it is not installed.

```
[3]: !pip install pinecone
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pinecone in ~/.local/lib/python3.10/site-packages
(7.0.1)
Requirement already satisfied: certifi>=2019.11.17 in
/usr/local/lib/python3.10/site-packages (from pinecone) (2023.11.17)
Requirement already satisfied: pinecone-plugin-interface<0.0.8,>=0.0.7 in
~/.local/lib/python3.10/site-packages (from pinecone) (0.0.7)
Requirement already satisfied: python-dateutil>=2.5.3 in
/usr/local/lib/python3.10/site-packages (from pinecone) (2.8.2)
Requirement already satisfied: typing-extensions>=3.7.4 in
~/.local/lib/python3.10/site-packages (from pinecone) (4.14.1)
Requirement already satisfied: urllib3>=1.26.0 in
/usr/local/lib/python3.10/site-packages (from pinecone) (1.26.18)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.10/site-
packages (from python-dateutil>=2.5.3->pinecone) (1.16.0)
```

```
[4]: !pip install -qU \
    pinecone==7.0.1 \
    pinecone-notebooks
```

```
[9]: # Import libraries
from pinecone import Pinecone, ServerlessSpec
from transformers import CLIPModel, CLIPProcessor
from PIL import Image
import torch, json, glob
```

```
[6]: # Import initialize Pinecone and other libraries
# ---- CLIP (image embeddings) ----
model_id = "openai/clip-vit-base-patch32"
model = CLIPModel.from_pretrained(model_id)
processor = CLIPProcessor.from_pretrained(model_id)
```

```
/voc/work/.local/lib/python3.10/site-
packages/huggingface_hub/file_download.py:945: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
warnings.warn(
2025-08-23 18:16:48.028165: I tensorflow/core/util/port.cc:110] oneDNN custom
operations are on. You may see slightly different numerical results due to
floating-point round-off errors from different computation orders. To turn them
```

```

off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2025-08-23 18:17:21.456811: I tensorflow/core/platform/cpu_feature_guard.cc:182]
This TensorFlow binary is optimized to use available CPU instructions in
performance-critical operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other
operations, rebuild TensorFlow with the appropriate compiler flags.
2025-08-23 18:18:22.733261: I
tensorflow/compiler/xla/stream_executor/cuda/cuda_gpu_executor.cc:995]
successful NUMA node read from SysFS had negative value (-1), but there must be
at least one NUMA node, so returning NUMA node zero. See more at
https://github.com/torvalds/linux/blob/v6.0/Documentation/ABI/testing/sysfs-bus-
pci#L344-L355
2025-08-23 18:18:24.599439: I
tensorflow/compiler/xla/stream_executor/cuda/cuda_gpu_executor.cc:995]
successful NUMA node read from SysFS had negative value (-1), but there must be
at least one NUMA node, so returning NUMA node zero. See more at
https://github.com/torvalds/linux/blob/v6.0/Documentation/ABI/testing/sysfs-bus-
pci#L344-L355
2025-08-23 18:18:24.601781: I
tensorflow/compiler/xla/stream_executor/cuda/cuda_gpu_executor.cc:995]
successful NUMA node read from SysFS had negative value (-1), but there must be
at least one NUMA node, so returning NUMA node zero. See more at
https://github.com/torvalds/linux/blob/v6.0/Documentation/ABI/testing/sysfs-bus-
pci#L344-L355

VOC-NOTICE: GPU memory for this assignment is capped at 2048MiB

/voc/work/.local/lib/python3.10/site-
packages/huggingface_hub/file_download.py:945: FutureWarning: `resume_download`
is deprecated and will be removed in version 1.0.0. Downloads always resume when
possible. If you want to force a new download, use `force_download=True`.
warnings.warn(

```

```

[7]: def get_image_embedding(path):
    img = Image.open(path).convert("RGB")
    inputs = processor(images=img, return_tensors="pt")
    with torch.no_grad():
        emb = model.get_image_features(**inputs)           # [1, 512]
        emb = emb / emb.norm(p=2, dim=-1, keepdim=True)   # normalize
    return emb.squeeze(0).cpu().numpy().tolist()          # list[512]

```

```

[10]: # ---- Pinecone v3 client ----
pc = Pinecone(api_key="psk_67VPttt4Ey")

index_name = "image-index"
dimension = 512 # CLIP ViT-B/32 output size
metric = "cosine"

```

```

# create index if it doesn't exist
existing = [i.name for i in pc.list_indexes()]
if index_name not in existing:
    pc.create_index(
        name=index_name,
        dimension=dimension,
        metric=metric,
        spec=ServerlessSpec(cloud="aws", region="us-east-1") # change region
        ↪as you like
    )

index = pc.Index(index_name)

```

```

[11]: # ---- upsert multiple images (e.g., up to 50 from ./images) ----
image_paths = glob.glob("./images/*.jpg")[:50] + glob.glob("./images/*.png")[:
    ↪50]

vectors = []
for i, p in enumerate(image_paths):
    vec = get_image_embedding(p)
    vectors.append({"id": f"img-{i}", "values": vec, "metadata": {"filename":
    ↪p}})

# upsert to a namespace (optional)
index.upsert(vectors=vectors, namespace="images")
print(f"Upserted {len(vectors)} vectors.")

```

Upserted 50 vectors.

```

[12]: # ---- similarity search with a query image ----
query_path = image_paths[0] # or any other image path
qvec = get_image_embedding(query_path)

res = index.query(
    vector=qvec,
    top_k=3,
    include_metadata=True,
    namespace="images"
)

# pretty print (v3 objects support .to_dict())
print(json.dumps(res.to_dict(), indent=2))

```

```

{
  "matches": [],
  "namespace": "images",
  "usage": {

```

```

        "read_units": 1
    }
}

```

```

[13]: # ---- similarity search with a query image ----
query_path = image_paths[6] # or any other image path
print(query_path)
qvec = get_image_embedding(query_path)

res = index.query(
    vector=qvec,
    top_k=3,
    include_metadata=True,
    namespace="images"
)

# pretty print (v3 objects support .to_dict())
print(json.dumps(res.to_dict(), indent=2))

```

```

./images/45.jpg
{
  "matches": [
    {
      "id": "img-6",
      "score": 1.00073242,
      "values": [],
      "metadata": {
        "filename": "./images/45.jpg"
      }
    },
    {
      "id": "img-13",
      "score": 0.715362549,
      "values": [],
      "metadata": {
        "filename": "./images/12.jpg"
      }
    },
    {
      "id": "img-40",
      "score": 0.674179137,
      "values": [],
      "metadata": {
        "filename": "./images/22.jpg"
      }
    }
  ],
  "namespace": "images",

```

```

    "usage": {
        "read_units": 1
    }
}

```

[14]: *#show both the query image and the top-K retrieved images in a grid with ↵*
↳matplotlib.

```

import matplotlib.pyplot as plt

def show_query_and_results(query_path, results, k=5):
    """
    Display the query image + top-K results from Pinecone in a grid.
    """
    # load query image
    query_img = Image.open(query_path).convert("RGB")

    # plot query image
    plt.figure(figsize=(12, 6))
    plt.subplot(1, k+1, 1)
    plt.imshow(query_img)
    plt.title("Query")
    plt.axis("off")

    # plot retrieved images
    for i, match in enumerate(results['matches'][:k], start=2):
        fname = match['metadata']['filename']
        img = Image.open(fname).convert("RGB")
        plt.subplot(1, k+1, i)
        plt.imshow(img)
        plt.title(f"Score: {match['score']:.2f}")
        plt.axis("off")

    plt.tight_layout()
    plt.show()

```

[15]: *query_path = image_paths[4] # or any other image path*
show_query_and_results(query_path, res.to_dict(), k=3)

Query



Score: 1.00



Score: 0.72



Score: 0.67



```
[16]: pc.delete_index(name=index_name)
```

```
[ ]:
```